



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid to: April 30, 2018

Certificate Number: 1876.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Dimensional

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Calipers, Verniers and Linear Scales ³	Up to 36 in	(43 + 2.4L) μin	Gage blocks, rod standards
Height Gages ³	Up to 36 in	(33 + 2.7L) μin	Gage blocks
Dial and Test Indicators ³	Up to 2 in	160 μin	Gage blocks
Micrometers ³ –			
Outside	Up to 36 in	(28 + 2.6L) μin	Gage blocks, rod standards
Inside	Up to 36 in	(28 + 2.6L) μin	
Depth	Up to 36 in	(28 + 2.6L) μin	
Pin Gages ³	(0.011 to 1.00) in	84 μin	Digital micrometer

Parameter/Equipment	Range	CMC ^{2, 4} ($\pm$)	Comments
Gage Blocks	(0.3125 to 4) in	$(2.6 + 1.3L) \mu\text{in}$	Gage block comparator
Rules ³ and Tapes ³	Up to 360 in	$(0.058 + 9.3 \times 10^{-6}L) \text{ in}$	Gage blocks, linear standards
Thread Plugs – Standard 60° Screw, Metric Screw			
Major Diameter	Up to 6 in	$(18 + 3.2L) \mu\text{in}$	Supermicrometer TM with thread measuring wires
Pitch Diameter	Up to 6 in	170 μin	
Thread Plugs – Pipe Inch (NPT, NPSM, NPSL, ANPT, NPTF)			
Major Diameter	Up to 6 in	$(20 + 3.3L) \mu\text{in}$	Supermicrometer TM with thread measuring wires
Pitch Diameter	Up to 6 in	180 μin	
Surface Plates ³ – Repeatability Only	18 in $\times$ 24 in to 72 in $\times$ 96 in	50 μin	Repeat-o-meter

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 6} ($\pm$)	Comments
DC Voltage – Generate ^{3, 5}	(0 to 330) mV 330 mV to 3.3 V (3.3 to 33) V (30 to 330) V (100 to 1000) V	61 $\mu\text{V/V} + 3 \mu\text{V}$ 51 $\mu\text{V/V} + 5 \mu\text{V}$ 52 $\mu\text{V/V} + 50 \mu\text{V}$ 65 $\mu\text{V/V} + 500 \mu\text{V}$ 64 $\mu\text{V/V} + 1.5 \text{ mV}$	Fluke 5500A
DC Voltage – Measure ³	(0 to 100) mV 100 mV to 1 V (1 to 10) V (10 to 100) V (100 to 1000) V (1000 to 40 000) V	50 $\mu\text{V/V} + 0.35 \mu\text{V}$ 40 $\mu\text{V/V} + 0.7 \mu\text{V}$ 37 $\mu\text{V/V} + 0.5 \mu\text{V}$ 47 $\mu\text{V/V} + 0.6 \mu\text{V}$ 47 $\mu\text{V/V} + 10 \mu\text{V}$ 2.3 % rdg	Agilent 34401A

Parameter/Equipment	Range	CMC ^{2, 6, 7} ($\pm$)	Comments
DC Current – Generate ⁵	(0 to 3.3) mA (3.3 to 33) mA (33 to 330) mA 330 mA to 2.2 A (2.2 to 11) A (11 to 500) A	0.013 % + 0.05 μ A 0.01 % + 0.25 μ A 0.01 % + 3.4 μ A 0.034 % + 44 μ A 0.065 % + 0.33 mA 0.041 %	Fluke 5500A Fluke 5500A plus coil, clamp on only
DC Current – Measure	(0 to 10) mA (10 to 100) mA 100 mA to 1 A (1 to 3) A (3 to 20) A	0.05 % rdg + 0.02 % rng 0.06 % rdg + 0.005 % rng 0.1 % rdg + 0.01 % rng 0.1 % rdg + 0.01 % rng 0.047 % rdg + 0.02 % rng	Agilent 34401A Fluke 8508A
Resistance – Generate ^{3, 5}	(0 to 11) Ω (11 to 33) Ω (33 to 110) Ω (110 to 330) Ω (0.33 to 1.1) k Ω (1.1 to 3.3) k Ω (3.3 to 11) k Ω (11 to 33) k Ω (33 to 110) k Ω (110 to 330) k Ω (0.33 to 1.1) M Ω (1.1 to 3.3) M Ω (3.3 to 11) M Ω (11 to 33) M Ω (33 to 110) M Ω (110 to 330) M Ω	0.013 % + 0.008 Ω 0.013 % + 0.015 Ω 0.01 % + 0.008 Ω 0.01 % + 0.008 Ω 0.093 % + 0.06 Ω 0.093 % + 0.06 Ω 0.093 % + 0.6 Ω 0.093 % + 0.6 Ω 0.012 % + 6 Ω 0.013 % + 6 Ω 0.016 % + 55 Ω 0.016 % + 55 Ω 0.061 % + 550 Ω 0.11 % + 550 Ω 0.5 % + 5.5 k Ω 0.5 % + 17 k Ω	Fluke 5500A
Resistance – Measure ³	(0 to 100) Ω 100 Ω to 1 k Ω (1 to 10) k Ω (10 to 100) k Ω 100 k Ω to 1 M Ω (1 to 10) M Ω (10 to 100) M Ω 1 k Ω to 1 M Ω	0.01 % rdg + 0.004 % rng 0.01 % rdg + 0.001 % rng 0.01 % rdg + 0.001 % rng 0.01 % rdg + 0.001 % rng 0.01 % rdg + 0.001 % rng 0.04 % rdg + 0.001 % rng 0.8 % rdg + 0.01 % rng 0.01 % rdg + 0.001 % rng	Agilent 34401A
Fixed Points	100 Ω 10 M Ω 100 M Ω	0.01 % rdg + 0.004 % rng 0.04 % rdg + 0.001 % rng 0.8 % rdg + 0.01 % rng	

Parameter/Range	Frequency	CMC ^{2, 6} ($\pm$)	Comments
Capacitance – Generate ⁵	(0.33 to 0.5) nF (0.5 to 1.1) nF (1.1 to 3.3) nF (3.3 to 11) nF (11 to 33) nF (33 to 110) nF (110 to 330) nF (0.33 to 1.1) μ F (1.1 to 3.3) μ F (3.3 to 11) μ F (11 to 33) μ F (33 to 110) μ F (110 to 330) μ F (0.33 to 1.1) mF	0.51 % + 0.01 nF 0.51 % + 0.01 nF 0.51 % + 0.01 nF 0.51 % + 0.01 nF 0.27 % + 0.1 nF 0.27 % + 0.1 nF 0.27 % + 0.3 nF 0.27 % + 1 nF 0.38 % + 3 nF 0.38 % + 10 nF 0.45 % + 30 nF 0.52 % + 100 nF 0.74 % + 300 nF 0.3 % + 300 nF	Fluke 5500A
AC Voltage – Measure ³			
(1 to 100) mV	(5 to 10) Hz 10 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz	0.35 % rdg + 0.04 % rng 0.06 % rdg + 0.04 % rng 0.12 % rdg + 0.05 % rng 0.6 % rdg + 0.08 % rng 4 % rdg + 0.5 % rng	Agilent 34401A
(1 to 750) V	(5 to 10) Hz 10 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz	35 % rdg + 0.03 % rng 0.06 % rdg + 0.03 % rng 0.12 % rdg + 0.05 % rng 0.6 % rdg + 0.08 % rng 4 % rdg + 0.5 % rng	
750 V to 40 kV	60 Hz	2.3 % rdg	

Parameter/Range	Frequency	CMC ^{2, 6} ($\pm$)	Comments
AC Voltage – Generate ^{3, 5}			
(1 to 33) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.35 % + 20 μ V 0.15 % + 20 μ V 0.2 % + 20 μ V 0.25 % + 20 μ V 0.35 % + 20 μ V 1.0 % + 20 μ V	Fluke 5500A
(33 to 330) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.25 % + 50 μ V 0.05 % + 20 μ V 0.1 % + 20 μ V 0.19 % + 40 μ V 0.24 % + 170 μ V 0.7 % + 330 μ V	
330 mV to 3.3 V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.15 % + 250 μ V 0.03 % + 60 μ V 0.08 % + 60 μ V 0.14 % + 300 μ V 0.24 % + 1.7 mV 0.5 % + 3.3 mV	
(3.3 to 33) V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.15 % + 2.5 mV 0.045 % + 0.6 mV 0.083 % + 2.6 mV 0.2 % + 5 mV 0.24 % + 17 mV	
(33 to 330) V	45 Hz to 1 kHz (1 to 10) kHz (10 to 20) kHz	0.21 % + 6.6 mV 0.22 % + 15 μ V 0.22 % + 33 μ V	
(330 to 1020) V	45 Hz to 1 kHz (1 to 10) kHz (5 to 10) kHz	0.21 % + 80 mV 0.28 % + 0.1 mV 0.29 % + 0.5 mV	

Parameter/Range	Frequency	CMC ^{2, 6, 7} ($\pm$)	Comments
AC Current – Generate ^{3, 5}			
(0 to 330) μ A	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.25 % + 0.15 μ A 0.13 % + 0.15 μ A 0.13 % + 0.25 μ A 0.4 % + 0.15 μ A 1.3 % + 0.15 μ A	Fluke 5500A
330 μ A to 3.3 mA	(10 to 20) Hz 20 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.2 % + 0.3 μ A 0.1 % + 0.3 μ A 0.1 % + 0.3 μ A 0.2 % + 0.3 μ A	
(3.3 to 33) mA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.2 % + 3 μ A 0.1 % + 3 μ A 0.1 % + 3 μ A 0.2 % + 3 μ A 0.6 % + 3 μ A	
(33 to 330) mA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.2 % + 30 μ A 0.1 % + 30 μ A 0.09 % + 30 μ A 0.2 % + 30 μ A 0.6 % + 30 μ A	
330 mA to 2.2 A	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz	0.2 % + 300 μ A 0.1 % + 300 μ A 0.8 % + 300 μ A	
(2.2 to 11) A	(45 to 65) Hz (65 to 500) Hz 500 Hz to 1 kHz	0.07 % + 2 mA 0.1 % + 2 mA 0.34 % + 2 mA	
(11 to 500) A	(50 to 60) Hz	0.39 %	Fluke 5500A plus coil, clamp-on only

Parameter/Range	Frequency	CMC ^{2, 6} (±)	Comments
AC Current – Measure ³			
(0 to 1) A	(3 to 5) Hz (5 to 10) Hz 10 Hz to 5 kHz	1.6 % rdg + 0.04 % rng 0.4 % rdg + 0.04 % rng 0.16 % rdg + 0.04 % rng	Agilent 34401A
(1 to 3) A	(3 to 5) Hz (5 to 10) Hz 10 Hz to 5 kHz	1.2 % rdg + 0.06 % rng 0.4 % rdg + 0.06 % rng 0.17 % rdg + 0.06 % rng	
(3 to 20) A	10 Hz to 2 kHz 2 kHz to 10 kHz	0.11 % rdg + 0.01 % rng 0.29 % rdg + 0.01 % rng	Fluke 8508A

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Calibration of Thermocouple Indicating Systems ^{3, 5} –			
Type E	-50 °C to 650 °C 650 °C to 1000 °C	0.29 °C 0.32 °C	Fluke 5500A
Type J	-210 °C to 150 °C 150 °C to 1200 °C	0.36 °C 0.33 °C	
Type K	-200 °C to -100 °C -100 °C to 120 °C 120 °C to 1372 °C	0.4 °C 0.35 °C 0.46 °C	
Type R	0 °C to 250 °C 250 °C to 1768 °C	0.61 °C 0.46 °C	
Type S	0 °C to 250 °C 250 °C to 1768 °C	0.61 °C 0.46 °C	
Type T	-200 °C to 0 °C 0 °C to 400 °C	0.33 °C 0.3 °C	

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Calibration of RTD Measuring Equipment ^{3, 5} –			
Pt 385, 100 Ω	-200 °C to -80 °C -80 °C to 0 °C 0 °C to 100 °C 100 °C to 300 °C 300 °C to 400 °C 400 °C to 630 °C 630 °C to 800 °C	0.14 °C 0.14 °C 0.15 °C 0.16 °C 0.17 °C 0.19 °C 0.37 °C	Fluke 5500A
Pt 385, 1000 Ω	-200 °C to -80 °C -80 °C to 0 °C 0 °C to 100 °C 100 °C to 260 °C 260 °C to 300 °C 300 °C to 400 °C 400 °C to 600 °C 600 °C to 630 °C	0.13 °C 0.13 °C 0.13 °C 0.14 °C 0.15 °C 0.15 °C 0.27 °C 0.37 °C	

III. Fluid Quantities

Parameter/Equipment	Range	CMC ² (±)	Comments
Volume ³ (Water)	(1 to 100) gpm	2.9 % rdg	Gravimetric determination
Air Velocity – Measure ³	(0.2 to 1) m/s (1 to 25.0) m/s	0.2 m/s 6 % rng + 0.1 m/s	Extech SDL350 anemometer

IV. Mechanical

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Scales ³ & Balances ³	1 mg to 32 kg Up to 20 000 lb	1.5 <i>LSD</i> 1.5 <i>LSD</i>	Verification with Class 1, 4 & F weights

	(20 000 to 80 000) lb (80 000 to 120 000) lb	2.3 <i>LSD</i> 3.4 <i>LSD</i>	
Parameter/Equipment	Range	CMC ^{2,4} (±)	Comments
Pressure Gauges ³ , Transducers ³ , and Transmitters ³	(-14 to 30) psi (10 to 16 000) psi	0.033 psi 0.009 % rdg	Pressure comparison
Torque Wrenches ³	(4 to 50) in·lbf (30 to 400) in·lbf (80 to 100) in·lbf (20 to 250) ft·lbf (200 to 2000) ft·lbf	0.15 in·lbf 1.2 in·lbf 2.9 in·lbf 0.73 ft·lbf 5.8 ft·lbf	Digital torque calibrator
Torque Analyzers ³ and Transducers ³	5 in·lbf to 2000 ft·lbf	0.2 % of applied load	Torque arms with Class 4 & F weights, including specific arms
Force – Tension and Compression ³	(1 to 1000) lbf (1001 to 10 000) lbf (10 001 to 100 000) lbf	0.029 % rdg 0.013 % rdg 0.013 % rdg	Class 4 & F weights and deadweights, ASTM E74 Load cell comparison, ASTM E74

V. Thermodynamics

Parameter/Equipment	Range	CMC ² (±)	Comments
Temperature Measure	-40 °C to 350 °C	0.06 °C	Thermometers, dry blocks, baths, ovens, temperature recorders
Relative Humidity – Measure ³	(0 to 80) % RH	2.3 % RH	Vaisala HMI41 w/ HMP42, salt chambers
Chart Recorders ³ and Data Loggers ³	-40 °C to 350 °C (10 to 90) % RH	2 % full scale temp 2.5 % RH	Temperature and humidity calibration; applies only to chart recorders and data loggers with remote sensors

VI. Time & Frequency

Parameter/Equipment	Range	CMC ^{2, 4} ($\pm$)	Comments
Digital/Mechanical Tachometer ³	(20 to 29 999) RPM	0.1 % of rdg ± 1 LSD	Direct reflective pickup tachometer
Timers ³	15 s to 24 hr	The greater of 0.05 % full scale (fs) or 1 LSD	Reference stopwatch
Stopwatches ³	15 s to 24 hr	0.01 % full scale + 0.5 s	NISTIME 32

¹This laboratory offers commercial calibration services and field calibration services.

²Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴In the statement of CMC, L is the length in inches and LSD stands for Least Significant Digit.

⁵Fluke 5500A CMCs are based upon the temperature the standard was calibrated ($t_{cal} \pm 5$ °C) and assuming the instrument is zeroed at least every seven days or when the ambient temperature changes more than 5 °C. For Resistance, a zero calibration is performed at least every 12 hours within ± 1 °C of use. For AC Current, CMCs are determined with the LCOMP off. CMCs are also based upon 1-year floor specifications. CMCs are expressed as either a fraction of the reading or as a fraction of the reading plus a fixed floor specification.

⁶The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMCs are expressed as either a specific value that covers the full range or as a fraction/percentage of the reading plus a fixed floor specification.

⁷In the statement of CMC, percentages are percentage of reading, unless otherwise indicated.



Accredited Laboratory

A2LA has accredited

ALABAMA SCALE & INSTRUMENT, INC d.b.a. ASI Calibration Lab – Alabama

Mobile, AL

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 19th day of April, 2016.

A handwritten signature in blue ink, likely belonging to the Senior Director of Quality and Communications.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 1876.01
Valid to April 30, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.